

Topographic Survey Report

Construction of Workmen Housing Complex (BQ Area) at OIL, Duliajan



CLIENT:

RITES Ltd.

Plot No. 144, Sector-44,

Gurugram (Haryana) -122003



Surveyors:

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For,
M/S AC GOGOI & ASSOCIATES

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1. GENERAL:

The detailed topographic survey of the proposed site for development of Workmen Housing and the vicinity has been carried out. The purpose of this topographic survey report is to provide a comprehensive overview of the survey conducted at the specified site location. The survey aims to capture and document the Two-dimensional features of the land, including its contours, natural features, and any existing structures. The objectives of the survey are to develop a topographic reference system to facilitate smooth execution of planning, design, for the development of infrastructure.

The following steps were adopted for carrying out the survey work:

- Jungle Cutting of the proposed site
- Establishment of Control Network using DGPS,
- Transfer of TBM to site,
- Establishing the coordinates of the proposed land boundaries & area,
- Picking-up of all topographical details, elevations, ground features and trees detail and
- Preparation and submission of survey drawing and report.

2. PROJECT OVERVIEW:

The site located in the Duliajan, Assam. The site is characterized by a mix of flat terrains, dense bushes and moderate slopes.

Survey Specifications

Specification	Details
Survey Date	June 2 - June 15, 2025
Survey Team	Consisting of 4 professional surveyors
Equipment Used	South Total Station, Emlid DGPS RS2+
Co-ordinate System	WGS 84/UTM Zone 46 N
Bench Mark	SAT 1000
	Near Jaloni Club X = 730244.904 Y= 3028496.496 Z =121.272M

3. SURVEY METHODOLOGY, FIXING OF BENCH MARK AND TRAVERSING:

At the beginning reconnaissance survey was carried out of the entire area to be surveyed

and along the periphery of the proposed area. Jungle cutting was carried out manually to carry out survey works. Then DGPS was established at reference bench mark SAT-1000 and shifting of Bench Mark to TBMs was performed using Emlid DGPS Rs2+, RTK capabilities (Real-Time Kinematic, a GPS correction technology for enhanced accuracy) for real-time data correction, observing it for 20 minutes at each Bench Marks (TBMs) with reference to the Bench Mark Sat-1000. Traverse points/TBMs were selected and fixed on pillars, culverts, roads/platform & other locations wherever available (where these could not be disturbed) during reconnaissance survey.

The traversing was carried for horizontal and vertical control of the area shown to the survey team. The closed traverse was run connecting the selected traverse/TBM points. The points were fixed at a distance not more than 250 m to obtain greater accuracy. Measuring the included angles between successive traverse/TBM points and measuring the horizontal distances by Total Station carried out the traverse. All angles, distances were measured twice on both faces to eliminate any manual and instrumental errors. The traverse was closed back to the initial station. The closing error was checked and found to be within permissible limit. The residual error was adjusted and final co-ordinates of TBMs were calculated and cross-checked with DGPS established points for accuracy.

After completion of fixing horizontal control points the fixing of vertical control points was taken up. The double levelling work was carried out with respect to the benchmarks established. An Auto level was used. The levelling work was closed at the originating point and checked for any closing error. The closing error was found to be within permissible limit. The residual error was adjusted.

Considering the control points/TBMs, topographic survey work was carried out using South Total station N6 in the area for picking up details & spot levels.

Permanent Reference Bench Mark, SAT 1000, was located near Zaloni Golf Club, Duliajan with details, X = 730244.904; Y = 3028496.496; Z = 121.272 M and was transferred to the TBMs as under:

The list of TBM's established at Proposed site is given below:

Sr.	Stat. No.	Easting	Northing	Elevation	Description
1	BM	730244.904	3028496.496	121.272 M	Near Zaloni Club Duliajan

Sr.	Stat. No.	Easting	Northing	Elevation	Description
2	TBM 1	729403.418	3029413.048	121.070 M	On Concrete Pillar
3	TBM 2	729391.093	3029340.715	120.943 M	On Concrete Pillar
4	TBM 3	729341.407	3029342.009	121.040 M	On Concrete Pillar
5	TBM 4	729214.300	3029444.073	120.718 M	On Footpath
6	TBM 5	729464.868	3029470.557	121.003 M	On Road
7	TBM 6	729203.376	3029334.313	121.198 M	On Culvert

4. TOPOGRAPHICAL SURVEY:

Detailed Topographical survey of the over ground features and/or natural features like road/footpath, fencing, gates, Nala / drains/culverts, power lines, telephone lines, electric lines with poles, transformers were made. Bridges / culverts were surveyed with X, Y and Z values on all of them. The survey covers the control points and all topographical features as follows:

- Buildings, hutment's, sheds, structures,
- Boundary/fencing features,
- Roads, tracks, footpaths, etc.,
- Drains (Kacha/ Pucca), culverts with their types & invert levels,
- Religious structures,
- Poles of electric and telephone lines,
- Water pipe line, fire hydrants
- Trees with their names, girth,
- Survey boundary

5. DATA PROCESSING:

All data from the total station and other field data's were down loaded regularly. The data was then suitably processed to form ground features and proper connectivity of line and other features. Collected data were processed using advanced GIS Global Mapper, E-Survey and CAD software. Quality checks were performed to ensure data accuracy and integrity.

6. TOPOGRAPHIC MAP

The processed data culminated in generating a comprehensive topographic map, presenting a representation of the site's terrain features. This map includes Topo Survey

Map & contour lines at 0.30-meter intervals, spot elevations, and annotations of significant natural and man-made features.

7. BOUNDARY COORDINATES OF PROPOSED COMPLEX

Sr.	Easting (X)	Northing (Y)
1	729224.1118	3029444.0610
2	729402.0490	3029462.6090
3	729401.1223	3029412.6336
4	729452.0064	3029410.9358
5	729446.8757	3029346.8075
6	729437.4412	3029338.0595
7	729335.4224	3029344.2592
8	729328.1980	3029338.3328
9	729328.1980	3029306.3650
10	729311.3459	3029293.6860
11	729211.0812	3029304.0554
12	729201.0155	3029315.5947

8. KEY FINDINGS OF TOPOGRAPHIC SURVEY:

The proposed location is predominantly covered with dense bushes/jungle with mix of flat terrains, moderate slopes having existing utility services, roads, footpaths, fencing, underground structures, foundations/plinth. It is also inferred that area is down with respect to existing public road.

9. PRESENTATION FOR FINAL DRAWINGS

All the drawings have been prepared depicting the details as specified. Information of the terrain, having X, Y & Z values have been presented.

10. LIST OF TREES – ENCLOSED AS ANNEXURE-A

11. SURVEY DRAWINGS – ENCLOSED AS ANNEXURE – B


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12. SITE PHOTOS



BM SAT-1000



TBM-1



TBM-2



TBM-3



TBM-4



TBM-5



TBM-6